

Module 4 Review

Permutations:

If we have *n options* and we want to *pick k* out of those, in a *certain order*, we can get the total number of ways of doing so by using the *Permutation formula*:

$$P(n, k) = \frac{n!}{(n - k)!}$$

Combinations:

In this case, the *order doesn't matter*:

$$C(n, k) = \frac{n!}{(n - k)! (k)!}$$

Probability:

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Independent events:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$

Binomial Distribution

- *p* : Probability of success
- *q* : Probability of failure; *q* = 1 – *p*
- *Bernoulli trial* : each repetition of an experiment involving only two outcomes
- *n* : Number of Bernoulli trials
- *r* : Number of successes

$$\Pr(X = r; n, p) = C(n, r)p^r(1 - p)^{(n-r)} = \frac{n!}{r!(n - r)!}p^r(1 - p)^{(n-r)}$$

Conditional Probability

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}$$

Baye's rule

$$P(\text{Hypothesis}|\text{Evidence}) = \frac{P(\text{Evidence}|\text{Hypothesis}) \times P(\text{Hypothesis})}{P(\text{Evidence})}$$

Expectation

$$E[X] = \mu = \sum_{X=x} xP(X = x)$$

Random Variable

- **Discrete Random Variable:** A random variable that takes on a finite or countably infinite number of values. E.g. Results of a fair die roll
- **Continuous Random Variable:** A random variable that takes on a noncountable infinite number of values. E.g. Weight of a randomly chosen person

Probability Mass Functions (PMF): The probability distribution for a *discrete random variable* is usually called a PMF.

Properties of PMF:

- $f(x) > 0$ for x in sample space and 0 otherwise.
- $\sum_x f(x) = 1$. In other words, the sum of the probabilities of all possible outcomes of a given experiment equals to 1.

Probability Density Functions (PDF): the probability distribution function of a *continuous random variable* X is called a PDF.

Properties of a PDF:

- $f(x) > 0$ for x in sample space and 0 otherwise.
- The *area under the curve* is equal to 1.

Cumulative Distribution Function (CDF): A cumulative distribution function (CDF), usually denoted $F(x)$, is a function that gives the probability that the random variable, X , is less than or equal to the value x .

$$F(x) = P(X \leq x)$$

Variance of a Random Variable

$$\sigma^2 = Var(X) = \sum (x_i - \mu)^2 f(x_i)$$

****We take each value of x , subtract the expected value, square that value and multiply that value by its probability. Then sum all of those values****

The **Standard Deviation (σ)** of a random variable is the *square root of Variance*.

Properties of Variance and Expectation

▪ $E(c) = c$	▪ $Var(c) = 0$
▪ $E(cX) = c E(X)$	▪ $Var(cX) = c^2 Var(X)$
▪ $E(X + c) = E(X) + c$	▪ $Var(X + c) = Var(X)$
▪ $E(X + Y) = E(X) + E(Y)$	▪ $Var(X + Y) = Var(X) + Var(Y)$

	<i>Note: X and Y need to be independent</i>
--	---

CLT for Sample Means

$\mu_x = \text{Mean of } X$

$\sigma_x = \text{Standard Deviation of } X$

$\bar{X} = \text{Oserved sample mean}$

$n = \text{sample size}$

The random variable $\bar{X}$ has a *z-score* associated with it which is given by:

$$z = \frac{\bar{x} - \mu_x}{\left(\frac{\sigma_x}{\sqrt{n}}\right)}$$

Here,

$\mu_x = \text{mean of } X \text{ and } \bar{X}$

$\frac{\sigma_x}{\sqrt{n}} = \sigma_{\bar{x}} = \text{Standard deviation of } \bar{X} \text{ or standard error of the mean}$

The *z-score* tells you *how many standard deviations away from the mean* a given observation is while a *z-table* shows the percentage or *probability of a value* being *less than a given z-score* in a standard normal distribution.

Confidence Interval

Confidence interval = sample mean $\pm$ margin of error

$$\text{Margin of Error} = z \times \frac{\text{population standard deviation}}{\sqrt{n}} = z \times \frac{\sigma}{\sqrt{n}}$$

A t-interval for the mean

When σ is unknown or when sample size is less than 30.

The unbiased estimate of the population standard deviation is represented using S .)

$$S = \sqrt{\sum_{i=1}^n \frac{1}{(n-1)} (X_i - \bar{X})^2}$$

Here, $(n-1)$ is called the “*Degrees of freedom*”.

$$\text{Margin of Error} = t \times \frac{\text{unbiased estimate of } \sigma}{\sqrt{n}} = t \times \frac{S}{\sqrt{n}}$$

Let's Practice

1. A password must be a **4-digit code** (digits 0–9). Digits **can repeat**. How many passwords contain **at least one repeated digit**?
2. A password consists of **3 letters** followed by **2 digits**. Letters must be **distinct**; digits may repeat. How many possible passwords?
3. You have **4 math books**, **3 science books**, and **2 history books**. Books of the **same subject must stay together**. How many ways can all 9 books be arranged?
4. In a class of 40 students, 24 students take math, 18 take physics, and 10 take both. If a student is chosen at random, what is the probability the student takes **neither** subject?
5. A student guesses on a **5-choice** question. If the student's answer is **correct**, what is the probability they guessed rather than knew the answer? (Assume the student knows the answer with probability 0.7, and guesses otherwise.)
6. You flip a fair coin 9 times. What is the probability you get **more heads than tails**?

7. A password generator picks a 6-character password using uppercase letters (A–Z).

Let X = number of vowels in the password. Find $E[X]$.

8. A fair die is rolled 100 times. Let $\bar{X}$ be the sample mean.

Find $P(3.2 < \bar{X} < 3.6)$.

9. You sample 20 exam scores and get $\bar{X} = 76, s = 10$.

Construct a **99% CI** for the class mean.

10. You want to estimate average commute time with margin of error ± 3 **minutes** at **95% confidence**, and σ is known to be 12 minutes. What sample size is required?